

Got a special conversation today. I'm joined by Ben Pulandian of BEP research. Ben is a specialist investor and analyst in the semiconductor world and the semiconductor supply chain. Of course, that is what powers AI. Ben, welcome to monetary matters. Thanks for having me, Jack. Excited to be here. Love your podcast. would love to dive in on some interesting topics and some things that your listeners care about. I'm really glad you're here too. I really like your work. You're very in the weeds on the semiconductor world. I want to start by asking kind of the proposed the bear argument. Ben, perhaps many people watching this, many institutional investors who are a little skeptical about AI, skeptical about semiconductors, they may think you know, that this is the repeat of the dot com bubble. And so my first question to you is Cisco was a very profitable company. The earnings and the growth experience in the dot com bubble was extreme, just like it is now in AI. Why isn't Nvidia Cisco? Why isn't this a replay of the dot com bubble? Thanks. A lot of people would bring up Cisco and overlay the Nvidia chart. The challenge with Cisco and comparing it to fiber, in my opinion, Things like that are sort of like a commodity. So when you talk about data comm or telecommunications, it's the idea of transmitting the data, right? So it just goes from point A to point B. With compute and NVIDIA, you have the chance of actually creating intelligence, something from nothing. Everybody wants intelligence. It's not a commodity that anyone else can make. And I think that is a discerning difference. You needed a company like Cisco to basically overbuild or global crossing to die on that hill, to bring that capacity and bandwidth to have the compute that we have today a little different. So here's I'll start where I disagree. I think intelligence is kind of a commodity like open AI creates geniuses and Anthropic creates digital geniuses. Sometimes some geniuses are better than others. Now the other one company pulled ahead and there are some are better at math, better at writing, okay? But that is kind of like, it is kind of a commodity. But here's where I agree with you that semiconductors are not a commodity. The challenge is people are just still using AI to do routine work like, hey, help me plan my trip. You know, obviously write software or how do I make this recipe? What's this bug that...

I took a picture of, right? The next inflection of AI, which we'll see coming and what I'll be writing about soon, is the intersection of artificial intelligence and the bigger sciences, mostly material science, biotech and other things. Once you start discovering new materials, new medicines, that impact. of doing something 10 years in the lab trying to figure out if you can model it with AI and the actual GPUs and computers and find drug targets, that is a big unlock. You compress 10 years of work into one year and people need to realize the speed up that you get with intelligence. So you're saying that your bull case is not Only 15% of the population is using AI every day to plan their trips and do routine tasks. And that 15% is gradually going to head to 70 or 80%. That is not your bull case. Your, your bull case is that mainly the intelligence is going to be advanced at the frontier that can do tasks that previously required millions, tens of millions, billions of dollars in human. intelligence, and now that can be done much more cheaply and at scale. That's what you're saying. Yeah. I mean, going back to what Jensen has been saying at GTC for a while from NVIDIA, the whole point of all this is, can you do your life's work in your lifetime? So when you look at R&D for medicine and people working in the labs, there's a lot of trial and error. If you can scale that with and compress the time to get to better patient outcomes with better options. That is the unlock. You couldn't do that with classical computing. You can only do that with high performance computing, larger clusters, more data, the more data points you throw into it, genomics, like organs, NHS data from different places of the world. Once you have multivariable data points, you can all of a sudden triangulate into things that you weren't able to do before and find that needle in the haystack to help that one patient. Yes. And that bull case for semiconductors and AI is something that I'm seriously entertained and I encourage my watching to entertain. I also think then that a short-term 12 to 18 month bull case is that a tremendous amount of capital, way more than has already been spent, is going to be spent on that goal with the

Anthropic opening. I basically trying to build a digital God or, you know, a digital IQ person that never sleeps and has, you know, it's a billion years of a 160 IQ person in five minutes. You can, you can do. And whether or not that is going to be achieved within a short period of time, so much capital is going to be spent and that money is going to go to NVIDIA, land research, ASML, the entire semiconductor supply chain. So I think that that's

something I want to see is that the in over the next 12 months, the bull case doesn't. need building the 100 IQ person in a data center to become true. It just needs the accelerated capex. That's what I would say. What's your reaction to that? Yes. And the parallel to that also is obviously the best, smartest, fastest AI from a national defense perspective is the best military. So if we step back and say, oh, this is all a waste and we're not going to do it, then. You have other countries or bad actor nations that could basically have the best AI and infiltrate all of our digital systems and basically cripple our economy and virtually take over our country. So people need to realize it's just not, again, chatbots. The race of the best AI is the best military. And the best military in the world basically sets policy around the world. And that's where I've written a lot about it in my thesis of the token dollar. Everything's denominated in tokens. How the world has enough oil. You can't close the straight of compute, right? It's only power constrained and the country that leads in the highest compute per watt at scale wins. And it's a race against us against China. So it's sort of like the space race reversed instead of Russia. It's China this time. Explain. very briefly what a token is and then why is there such a Shortage of tokens that you forecast for to further to be many many years and why won't there be an excess of tokens such as there was an excess of oil after that the the increase in the price of oil in 2022 there was a There was a surplus that lasted for many years and told the Iran War why isn't there going to be a shortage in tokens that ultimately leads to a glut you know the in most commodities and most things, shortages leads to gluts because there's overproduction. Are you saying that you don't think that's ever going to happen or it's going to be, you know, five, 10 years? It's unlikely. Explain what gives you that confidence in your view here. To draw a parallel to oil, all tokens aren't created equal. So when you start with a barrel of oil, it's either West Texas crude or Brent, Brent. And then from there, you refine kerosene and or gasoline or diesel.

And the way compute is heading based on the scaling laws, if you see from Anthropic and OpenAI, the more money and more compute they throw out these bigger models, they're getting better, more intelligent results. And the idea is, if you're doing compute, what model are you getting your intelligence from? If you're getting it from a free, I don't know, 30 billion parameter Lama model for Meta from a few years ago, like that's like. It's

like water at this point. It's like worthless. It barely does anything. But if you want frontier intelligence with the best model at the highest speed of answers and tokens, that's almost like jet fuel, right? The most expensive type of fuel. So you're going to see this segmentation in the market on different types of tokens and the speed and the intelligence that you get it at. And will there be a supply glut of intelligence? I don't know. What I know right now is that me, just one guy trying to use Fable 5 for the past weekend, I've hit my rate limits over and over again. And I had to buy more credits and compute. And it's a scary feeling when you're mid through a project and it says, sorry, come back in four hours before I can finish it. It's not good. I mean, if they're doing that to me, who else are they doing it to? And it's just another data point that they're just very compute constrained. Explain the difference, Ben, between the frontier model like Fable from Anthropic or opening-eyed news model that is going to come out in early July. What is the difference between what the frontier models can do and what the decent models, like a model from 12 months ago that was considered a cutting edge 12 months ago, can do? I think it just comes down to the data that it's trained on. Obviously, like there's cutoffs in intelligence. So some stuff is cut off because they close the books like, oh, this I'm only recent up until January of 2026. So the more history and time you put into it, obviously the bigger the model is. And then there's also things called world models that you can input audio and video. So the more multivariable it is and the more flexible it is. Obviously it provides more intelligence and then some of these models are tuned for specific tasks like agentic coding, software development, where basically you can turn English into computer language to build things in the virtual world or software. And the better the frontier models at these things, the better results you get and your code or your software is stronger and made with possibly less lines of code and less bloat.

So what percentage right now of the value and maybe of the consumption and spending on the frontier tokens is specifically in software engineering? From an enterprise standpoint, from what you gather from companies, almost all software developers are using agent decoding at this point. Because if you're running an enterprise, it's not, uh, it's not all you can eat. They're all running APIs. So it's consumption based and. That's where you see the huge inflection in Anthropic's ARR is they made a bet on the

enterprise. And I remember I was using last summer, my friend showed me Claude Sonnet. He's like, oh, I'm using Claude. I'm building all these things. I'm multiple windows open. And I remember I was trying to build a few things and Claude would tell me he would do it. And then I'm like, okay, where is it? He's like, oh, I'm sorry I lied. And I was like, dude, like, come on, like I spent hours coding and I didn't get anywhere. And then I let it go. And then in November, he's like, Hey, they came out with this thing called opus. It's much better. It doesn't hallucinate as much. And that really was the inflection point for anthropic. Once people realized that the coding model improved and it hallucinated less and it lied less. And obviously it has those agentic workflows where it's controlling parts of your computer from Xcode to other opening windows and looking at things. You saw the future in front of you. And it was pretty amazing. So you feel like You kind of have, we have definitive proof that in the world of coding, the advancements here are revolutionary and certainly you're going to generate a lot of revenue. What do you think the odds are though that it's limited to coding? The idea of coding is just one application. So the idea is to take this and how do you make like a cloud code for law? How do you make a cloud code for finance? How do you make a cloud code for science? How do you make a cloud code for education? How do you turn natural language into a tool to build things and learning? I think that's the idea and then run loops around that. There's a repetitive task in your organization. How can somebody who doesn't have a computing background build a script with cloud code or some coding assistant to automate that task so they don't have to waste their time on something mundane to work on something more productive, to grow the business, to grow their minds? I think that's the unlock for a lot of the AI stuff is Like, you know, stop trying to build spreadsheets that you've already built a template once, right? And investment banking analysts, sensitivity tables. That was like the hardest thing. Just give the inputs and have Claude make it. Go focus on something else. Focus on something more strategic. I think that's where I see the unlock and the higher productivity for enterprises and people. Now, tell us the role of...

semiconductors, the semiconductor stocks have absolutely been on fire. Just what is it about AI that demands so much semiconductors in a way that semiconductors have so much pricing power such as they have? I mean,

NVIDIA's revenues and profits have just been unbelievably, unbelievably high, you know, from in 2020. three to 27 billion to last 12 months, 250 billion. And an operating profit, you know, over the next 12 months, I think it's going to make 200 billion. That would be a conservative estimate, just looking at it made more than that in the last quarter. And by the way, none of that included any markups. I'm not talking about net income. I'm just talking about operating profit. So that is pre-tax. And by the way, Samsung reported, and they had a quarterly operating profit that was higher than Nvidia's because the squeeze in memories is so high. So I think that the semiconductor stocks have gone up, but the multiples have gone flat to down. This has not been at all a valuation increase. This has not been a bull market driven by valuation, the multiple going up. It's been the earnings going up and going up so much. So I just want to set the stage for you there. I think you have to really go back over 25 years. I graduated college in 04 right after the dotcom bubble of electrical engineering. I got a job in investment banking in San Francisco at Jeffree's Broadview. The guy that hired me, the associate Alvin Lin, the first day on the job is like, hey, actually, you're a great candidate, but I'm quitting. I'm going to go work for this company called NVIDIA. He's still there. I think he reports up to Jensen, great guy. And then at the same time, we actually had a mandate to sell this accelerated digital. media software company for mobile processors to NVIDIA. And I think they eventually absorbed it. It was called PaceSoft Silicon. And NVIDIA at that time was just a graphics computing company. There was a lot of hardware companies in Silicon Valley. I mean, Intel was a leader. And from 04 then on, it was sort of the birth of the new internet. That's when Facebook took off with AWS and Microsoft and Salesforce. So you've had this regime give or take of 20 years of classical based computing, which was just software as a service. So if you would go up and down Sand Hill Road or to any investor in New York, they're like, Oh, you want to raise 20 million to make a chip? Oh, that's too expensive. You got to like engineer it. And then you got to tape it out of TSMC and, you know, like it's just, it's just, I can put less money in this, this like.

B2B SaaS and I know I'm going to get, you know, 10 times ARR and it's going to be amazing. And that's why if maybe you or your readers or your listeners can actually chart maybe the Bessemer CLOU index or IGV over the last 20 years, kind of see the multiples and how it's gone up a lot. And

obviously there's a lot of stock based compensation or rule of 40 and like all this stuff that they would hype up into SaaS and ARR, right? The hardware guys, all the semiconductor guys for the last 20 years were just like, no one loved them. Like I remember in college, I had an internship at this company called Simer, Bob Akins was the founder CEO. He went to UC San Diego, him and his partner, Rick Sandsorm, developed the DUV XMR laser, right? So our customers then at Simer were obviously ASML, Nikon, and The end users were TSMC Intel and other fabs around the world. And he was like, at that time, the market cap was only a billion dollars. And he was, he was like stuck in small cap hell. And he was like struggling. And eventually they sold the company to ASML. So without that laser and ASML making the fabrication equipment, we wouldn't be here today. So, you know, going full circle to what I'm saying is there's been 20 years of lack of investment in the hardware capex cycle. And all we've been doing the last 20 years is building regular data centers with Intel CPUs or AMD CPUs or whatever Amazon's doing with their CPU chip and just hosting email and, you know, some browsing and web pages. And maybe the only company that was actually doing AI at the moment was Google with their TPU program, optimizing stuff for AlphaGo and search and recommendations. And then all of a sudden, I mean, I invested in NVIDIA in the fall of 2016 when DGX came out and all that other stuff. Not bad. Not bad, by the way. At that time, they had the advertiser on Facebook. It was like, power your machine learning or your AI journey. It was like a 50,000 box. I have the photo. It was like a Facebook ad. And I was like, yeah, machine learning, that makes sense. In fall of 2022 when the chat GPT app came out and everyone down and was like, holy shit. Like, it's actually pretty smart to answer my question. It's not stupid. And then that was the inflection iPhone moment for for AI. And I remember, you know, my 2007 iPhone moment, I remember when Steve Jobs showed it and I went to the AT&T store and I was like, I'm going to get a blackberry world because I'm a businessman. Businessmen use blackberries when I saw the iPhone.

And I was like all of glass. And it was like beautiful. I was like, holy shit. This is like amazing. And like that was the iPhone moment. No one looked back. What's Blackberry? I mean, what's Motorola? Like everyone forgot about that stuff. So then you basically, the iPhone moment for 2007 until

now was a huge supply chain shock. in itself for baseband processors, memory, CMOS image sensors for the camera, capacitors. They went through all that at Cupertino with Tim Cook. And they basically, the supply chain has matured. So fast forward to 2015 years later with this NVIDIA, was it H100 at the time? That actually made all this stuff. And I was like, oh, my God, this stuff actually works. You can parallelize intelligence. the more GPUs you put, the better answers you get. And then like the AI boom took off. So we're four years into that this November, okay? Four years. And it's not as easy as going to Taiwan and China and build a supply chain and come out with a new phone every year and have consumers buy it. You have basically like the largest companies in the world, basically as your buyer. And then you have to go get a piece of land, permit it, make sure there's power. get a bunch of tradesmen from all across the United States that are very limited, build it, energize it, make sure it works, and bring it online. So the two are not the same. And that's why you see this lag and this huge investment in semiconductor CapEx, regular semiconductor stocks, and obviously NVIDIA because There's been this 20-year lag in investment and care for whatever it's been doing, because all the love and attention has been going to capital-like businesses like software. And when you say the two are not the same, what are you comparing to? I think I have an idea, but I just want to be clear. I mean, I'm just drawing the parallel between Apple, but they're kind of like the same, but they're not. But it's the same sort of inflection. When you're using as an inflection point for iPhone moment, I think The idea is, the idea of a smartphone was always there with like a web browser and like all this other stuff, but like it was too clunky to use. I remember when I was at UC San Diego, I was part of the Qualcomm, Kiosera, they paid a testing program and they gave me this, this like brick phone that had a web browser. And I was like, look, I can check my emails, but it was like, it was this clunky crappy thing. It was like, it had no user experience, no user feeling, no soul.

But when the iPhone came out, it made sense. Like anybody can use it, saving a chat GPT. Anybody can be a data scientist. Anybody can ask a question. That's the idea is like the technologies are all out there. But what is the unlock to open it up to everybody? And that is that moment. And I want to compare CPUs versus GPUs. CPUs are sequential and GPUs that parallel processing. So they're running. on multiple nodes at the same time.



And is that why there is such a supply squeeze in the semiconductor supply chain? Is that GPUs are just so much harder to make? Or are there other things? In other words, why is there such a shortage of chips right now in a way that in most of the time, let's say, okay, memory prices or chip prices go up in 1994, They went down in 1995. Most of the time they resolve quickly. What is it about, about the specific semiconductor supply chain, about why it is so hard? Because honestly Ben, the reality is, and I, you know, I love commodities, but oil, like there's a secret, a secret about oil is that in Saudi Arabia, like you stick a stick in the ground and oil comes out. Like there's a lot of oil in the world. And of course there are shortages and straight up her moves. But you know, if the price of oil goes to 200, extremely intelligent. petroleum engineers around the world are going to work extremely hard to try and get oil out of very difficult to get places, and they're going to succeed. And the same is true over a long-term time horizon about semiconductors. But specifically, what about the semiconductor chain is so hard? Over the past three years, looking back and then over the next three years looking forward. NVIDIA itself is sort of like a very special company, because it's not strictly a semiconductor company. I think more than half the company is in the software space. And there's a lot of researchers and developers. They have their own open source frontier-ish type of model, Nemotron. And the stuff that I've written about is like, they can squeeze more performance per watt out of the existing chips just based on software optimizations. Okay. This is like the same thing as getting an over-the-air iOS update on your phone over the over-the-air update on your Tesla, right? Yeah, by the way, there were times I saw And like news that Tesla had to do a product recall. I'm like, oh my God, this is so bearish for the company. But then I realized it literally is just a software upgrade. Like they're not going to have to send the trucks back. Right. So any hardware company that obviously doesn't have like a software layer that provides the customer benefit over time commoditizes. And obviously you can make that argument for a lot of parts in the semiconductor supply chain. Memory has always been like a commodity. That's the argument.

Companies like Micron trade out a multiple of book value. That was always a metric when I would talk to funds and people. But apparently this time it is different because high bandwidth memory is special. It's not standard. And you need all this memory because the more context you give your AI

or your brain, the more data points it has and the better answer you'll get. That's all it comes down to. It's just having a bigger brain and remembering more things. I mean, that's why you see this huge inflection and memory all of a sudden, this huge demand. For years, it's been stable. And going back to CPU, yeah, because we were just running classical computing. Every year, every three years, you had to replace your computer, right? That was the corporate upgrade cycle for desktops. Oh, okay, it's been three years. We're gonna have a corporate upgrade refresh cycle. All that stuff's out the window now. Everything is changing. The dynamics are changing. And I think, you know, in video, it's like, because they have the software hardware code design, and then they work obviously with Anthropic and OpenAI and others, like they're tuning the software and hardware exactly to how these models work to the spec. And once you do that, you can squeeze out more performance. And I think that's where a lot of people are missing those things. And there's not an actual GPU shortage anymore. There was at some point, but I think I read the number. I think NVIDIA spent 110 billion on their supply chain, basically buying everything up for, I don't know, how many years out to make sure that, say, for example, one screw or one cable wire or whatever it is, if that's missing and the rack is like a million parts, that means your whole data center deployment is delayed. And every day, it's not a GPU shortage anymore than how come NVIDIA has such pricing power. And as does to a lesser degree, the other GPU companies AMD comes to mind, if there's not a GPU shortage. I think there's a shortage of powered land and tradesmen to build data centers as fast. I think that's where the disconnect is. People don't, again, you're not building millions of phones and selling them to consumers at AT&T stores and Apple stores. Like this is infrastructure. Infrastructure and buildings take time. You have neighbors petitioning. You have Sequa. You have water issues. Like infrastructure takes time and there's a lag, right? And sequencing our lag. So a shortage of infrastructure and powered land, but there's not a shortage of GPUs. Why is that a...

bullish scenario for Nvidia. Nvidia, by the way, actually on a forward multiple trades at a discount to its past five or 10 years, we can put up a chart of that. So the market is kind of pricing in that, that lack of pricing power or that their pricing power is going to go from extremely high to very high. It's still going to be very high, but just less so. And you know, the

gross margin is going to go down. The operating margin is going to go down. Why do you think that is or is not going to happen? I think the bearish argument with Nvidia is that Obviously, every large hyperscaler from Google to Amazon, Meta, was it even ByteDance? Everyone's going to make their own custom accelerator to try to not pay the Nvidia tax. Obviously, you're paying a higher price because you're getting a dependable product that ships on time with the software stack on top of it. But maybe if you can use it as a way to leverage on pricing or Maybe we'll get better performance. The jury is still out if you're actually saving money on doing it yourself. I think the only company that really has full use of their internal silicon is Google and their TPU. But they also buy billions of dollars a year of Nvidia GPUs and racks and serve it on the Google Cloud. And they make money on that too because there's so much demand. So either way, there's more demand for compute. then supply and it's just not the chips. It's just energized ready to go compute. Because like I said, getting the data centers up is the lag. It's not making these in factories or like everything at Taiwan Semiconductor is like fully automated to the max. You don't have to worry about that. Like they're good. All these chip companies are good. The lag is deploying it in the real world. Okay. Ben, so I'm going to venture and assume that that you own a lot more semiconductor stocks, you know, notionally in terms of value, then you do the infrastructure companies like iron or Corweave. I mean, prologus is, you know, something like that. Or this power suppliers eaten vertebra. Okay. And by the way, that is true about myself. But if what you said is true that the true shortage is in power land and not in chips. Does our portfolio positioning make sense? Why are we so long semiconductors if the true shortage is in powered land, not chips, other than memory? Because I think the backlog is measured in years and not quarters. So like I said, if a macro investor doesn't want to, is like semiconductors, it's just scared of semiconductors because of the whole cyclicity and definitely the way to really play this is power and infrastructure. Power is everything. If you can generate power.

profitably or have the infrastructure for it, and there's value today. And I think, you know, I've been writing about one of the most interesting companies that I think since December is bloom energy. They've been around for 20 years, gone nowhere. And the guy wanted to like power

houses with like this small fuel cell box that takes natural gas and converts it to electricity. And now, I think in less than like 80 days, they can deploy modular stuff at a data center site, hook up to the natural gas and produce electricity. And at 800 volt DC without all the the transformers and all this the crap in the middle of it. And it doesn't make any noise, right? So I think a lot of the challenge between with General Electric and Vernova with their turbines and like that stuff is noisy and weird. That's not the data center of the future. That's to me, that's like a temporary solution. because the grid is not enough capacity. We need to find long-term, better, sustainable power solutions because power is everything. And so so so many data centers that the grid doesn't support because it has to be approved by everything. So they want to go behind the meter. So, OK, we're going to turn natural gas into electricity. We're going to buy the turbines from the three companies that do that, you know, Mitsubishi. Uh, uh, g-venova and okay g-venova their backlog is out until 2028 2029. Okay. What else are we going to do? We're going to do turn this company that, you know, Ben Poole had been talking about called blue energy. That stock has done, uh, tremendously tremendously well. It's a little bit of a pullback and I actually don't know that that much about it, but why it seems to me like semiconductors are so much more difficult to do than this, this energy thing because Is it really that hard to turn natural gas into energy? I mean, there's even this guy, Boom Aviation, who's making a supersonic airplane. Instead of work focusing on the airplane, he's converting his whole manufacturing plant to making turbines because he thinks he can make turbines. Like everyone's running into this type of place. But I do think that sustainable, efficient energy generation is a net benefit for society and for these data centers and for the future. Obviously, you have guides like Elon saying, if we covered a part of Nevada with solar panels, it would be enough to power all the United States. It's possible. I think there's a combination of natural gas, solar, and batteries that I think will definitely be powering the future with some nuclear. Not SMRs, but just probably just traditional nuclear reactors, which is a clean energy source.

Because our electricity demand is just only going to go up. It's not going to go down. If we're going to add more electric cars, more electricity, more electronics, more electricity, more data centers, more GPUs, more electricity. So we're not going backwards anymore. We're only going

forwards. So Ben, the question that I asked of if the shortage is in the powered land, not the chips, why invest in the chips? Shouldn't you invest in the powered land? Famous now hedge fund manager, Leopold Ashtonbrenner has done with situational awareness. I don't think there are, you know, there are not as many chip stocks nearly as there are powered land stocks like iron or blue, blue energy. Do I ask you for a pushback to that? I don't, it sounds like you're not giving that much of a pushback. So maybe is that sounds like you think that that actually is wise? You have to kind of really look at the assets of each powered land player and at the end of the day, powered land is a quintessential real estate play. The powered land needs a quality tenant and that quality tenant needs to be paying a good price. So if you can line all those things up together and you think all those things will work out, then it'll be a great investment. But at the end of the day, it's a simple landlord tenant real estate play, which could be lower on the risk curve. But the question is, there's a bifurcation in how

Powered land players are playing. I think I went to the nebius event. They're buying different software companies. They're trying to build a full stack different solutions They're trying to be like the hyperscaler versus other guys who were like doing Bitcoin before the Bitcoin market is dead. Who's doing Bitcoin? Yeah, the iron name Ben I just struggle when you use a term like less risky I mean some of these companies are borrowing billions of dollars to build out physical infrastructure as opposed to and video, which is actually going to be borrowing like 20 billion for liquidity purposes, but it's just printing so much cash or even companies like ASML or lamb research that optically are a very high valuation, but are just so critical and their, their, uh, the market position is so dominant. It's all oligopolistic. Or in the case of the EUV for ASML, literally a monopoly that I, I think that those sound a little bit less risky than borrowing. 20 billion to build data centers in, in Texas. There is a lot of risk of that. And I think the, do you know that scene from the movie Batman where Bane is in the plane and then he looks at the guy is like one of us has to be dying the wreckage brother. So figure there's a bunch of neoclouds on that airplane. A few of them are going to die in the wreckage or get absorbed. I don't know who they are, but the whole idea of. of a neocloud and obviously a high leverage. I think what Fermi is one of them. There's a crazy IPO and the CEO did all these weird things and they have power of land and it's a

ground lease. I got pitched it a few times. It was just way over my head. I think from an investing standpoint, kiss, keep it simple, stupid, sometimes it's easier. But if you really want to, you have an itch for power of land or data centers, maybe buy utility, maybe... by Blackstone, I don't know. Blackstone, okay. No one is really talking about that. Blackstone's position in the data center market is very large. And that is there. Okay. What do you think about the, um, the, the powered land names, like Texas Pacific land or land bridge, which owns land out in West Texas, where data centers may or may not be developed. And in the case of TPL, uh, former executive of. Google is trying quite hard to build a data center there. What do you think of that? I mean, those things, just like I said, it's a quintessential real estate development play, high risk, high reward. Real estate in itself as an asset is filled with debt. I mean, it's a leveraged asset. So if you want to put leverage on leverage and hopefully get that awesome tenant and win and make it a home run and go in that direction.

For me, sometimes it's a little bit too hard to quantify because I don't know what's going on in that market, specifically who's there. I'm not looking at every county report. There's a lot of guys analyzing what things are filed. I mean, I even had this guy put together a bunch of land in Northern California. He's like, this land will be amazing for a data center. But the funny thing is, I'm like, how are you powering it? He's like, oh, we're powering of a bunch of blue energy. So I was like, oh, that's an interesting data point. So I think that to me was like, okay, another data point for blue energy. So like people are using it in these different places. Like I rather just invest in that. We'll use it for me. So you've been an investor in blue energy. Yeah. It's it to me, it's obviously misunderstood 20 years, no product market fit. And all of a sudden it became like the killer app because it's quiet energy. for the data center and it chips fast, right? Time to power is everything. So if you're a data center developer and you're borrowing at seven, eight percent, you want to wait for this power thing to come, or do you want to wait for the utility to finally hook you up when you have the GPUs and everything ready to go? Or are you going to be like, you know what? When Oracle is doing like, give me the bloom. Let's be online. Let's energize this project. Let's start charging the customer and let's move on. So that's. That's the way I look at it. Yes. For Texas Pacific land and land bridge, they probably not going to do the development. People would be

building it on their land. So less debt, less capex that they would be doing. And for full disclosure, I have been an investor in TPL. I actually sold my entire position as I record this. But that may or may change. I may reenter for, for, for various reasons. Just want to. Great. I have no position. Yes. Yes. Ben, Ben, have you heard of, I don't know, this series energy or series at C E R E S. They're trying to complete, compete with Bloom Energy. Uh, no, I have not. I think though one that's kind of run up recently is fuel cell or F cell. That's like an older company, which basically it's like the same type of technology fuel cells. I guess they're getting business. I haven't like. So in the chip world, Ben, how are you allocating your capital to invest in semiconductors and share your philosophy behind that? All chips aren't created equal. I think people need to really understand that. There's obviously memory chips. There's FPGAs. There's ASICs. There's even diodes. Remember, I started a commercial lighting company from 2005 to 2019. An LED diode, a light.

is like the dumbest type of LED you could use. It just off or on. It's light, right? And you can go all the way up to the harder stuff, which is obviously like a GPU, which is processing massive computational loads and is creating intelligence. So you got to pick where you want to go. But I think obviously you have players like Nvidia and then you have their MI series. I think they have an AI event in a few weeks in San Francisco. They want a piece of the market. Intel is trying again with investments in Samba Nova and other companies. But when you have a whole host of startups like etched or positron or unconventional AI, just like everyone's like, everyone sees like a 5 trillion market cap. And it's like every entrepreneur's dream or investor's dream is like, let's get a crazy team together. We're all smart. Oh, we're ex Google. or XN video, whatever it is. And let's just build a better chip than they can. And if we can just get 1% market share will be successful. And then one of these other companies will just buy us because we'll be a nuisance. But can they really scale the supply chain? Do they have 100 billion to buy all this memory and fly to Taiwan and move all these suppliers and cables and racks and like, like, have President Trump on speed dial and like, It's just very multifaceted. It's a rich man's game. And it's hard to pick smaller companies to think that they can really grow in. But there's different parts of the stack that win. I think one of the most interesting parts for me is obviously optics has been interesting with

communication and lasers and moving the data faster between GPUs. And then there's been this big push in, it's called CXL, which is memory pooling. The idea is that because memory prices are so high, instead of buying more memory, you kind of pool your memory in one place and you share it and you process it at a central location. So companies like Astera Labs was like left for dead and then all of a sudden shot up 400% in the same thing with Credo and Marvell, which focus on that space as well. So, but these are all like inflections in moments in time, you know, holding Nvidia for 10 years coming up in November, I've been through a lot of drawdowns. The big crypto drawdown in 2018, and then the 2020 boom in the 2022. I remember every day it was going down. These things aren't easy to hold. Everyone thinks it's glorious, but you just have to pay attention to the company, their long-term roadmap, management. What are they saying? Are they being honest? Are they doing what they're saying?

just look at how things are executing. Obviously the next, when I was at GTC in March, all this stuff is being built for this huge token explosion. And I think the next big explosion in tokens is physical AI and edge AI, which is basically robotics, doing agentic work, right? So like, you can think of robotics or robots as like when you're running Claude code or Claude, the CPU is telling these agents to do certain things. Orchestrating? So what if the CPU is orchestrating physical agents, which will be robots? That's the way it's headed. And you're telling the robot a sequence of things to do on the factory floor or a sequence to do in the lab makes these chemicals, right? It's sequencing in parallel with tokens and a loop. I mean, that's basically what it is. So Ben, if you're number one choice for a stock or company in the AI supply chain is what? It's Nvidia. I mean, I knew you'd say that. Okay. What is what is number two? I mean, I like Apple really interesting. Tell me about that. Apple is always going to be the company that's going to bring useful AI to the world, right? In a way where it's hopefully humanistic and relevant to your everyday life, right? I think AI is not going to be some sort of product that's just going to be an upgrade to whatever we have. And You can already see, uh, I mean, the iPhone was already using AI. You can already sort your album based on faces that you've trained it with that it recognizes and you can create videos on I was with my friends in this country. That's, that's like simple on-device AI. You just have to just. I, um, actually I'll give a shout out to the compound. I was



listening to Josh Brown and Michael Batnick. I was listening to their podcast, I think yesterday about just how much money Apple is going to make in services from Okay, I'm a user and I'm using Claude on my phone and I'm paying Claude and paying Anthropic Apple is going to get 20% of that 15 20% of that in the same way they do that through Apple and that has a hundred billion dollar It's called services But really it's just taking a cut of a lot of it is taking a cut of you know money that of the apps that people buy on the app store And I think that is going to be a big market So even if Apple isn't an innovator in AI, which it very well could be I think that is going to be a driver for the stock That being said, Ben, I think Apple intelligence has been very overwhelming to me and my personal experience as a consumer. Yeah, it sucks. Yeah, I thought they had they have all the data and they want to keep it private. But for some reason, something hasn't clicked, which is weird. But hopefully with Google and whatever, doing the Gemini or even Anthropic, or people can use their own models, you can get some on-device AI that's like relevant to what you're doing in your life.

and your social context and it's private to you. I think that's, that's the most important. I don't, people need to realize like, even with meta, people, I was doing the photos yesterday with the Instagram app, like the thing like, we're basically voluntarily training their models of our own data. Does that matter to you or not? I mean, from a privacy perspective, some people really care. I mean, but you have to understand Apple's ethos from the beginning is your data belongs to you. We won't train on your data. And I think that over time gets important when you have photos and text messages and things like that. That should all belong to you. Shouldn't belong to any frontier model lab. Right. So you like Apple as a, is it with this AI exposure? Technically, though, it is not, you know, in the semiconductor supply chain, even though it does design its own chips and stuff. So if strictly sticking to like stocks that would be in the, you know, Vanax semiconductor ETF or the, the ice semi index or Philly's, whatever index, like. Pure semi stocks. What is your number two? That's a good one. I still think even though we're in this crazy memory is sort of constrained and the only U.S. company that really does memory is Micron. But I think Hynix is going to listen to United States. I still see three companies dominating and memory is obviously a very interesting name. It just trades very weirdly. So, but I think you need to have a small allocation to memory. What do you make of

the semi-cap equipment companies like ASML, KLA, LAM Research, Tokyo Electron, that applied materials that actually are totally necessary for memory to expand? And if Samsung and SK Hynix and Micron are going to expand their capacity, which they're trying to do. They need to buy so many machines, so many machines, and there's recurring revenue that lamb research, like 30% of the revenue is recurring revenue because they need to sell us. They need to replace the spare parts, which obviously, you know, they're investors who in, in like auto zone or something, they get that, but they should know that this is also true of lamb research and also to, to maintain the machine software revenue and the like. So, you know, if lamb research revenue is going to scale massively, which I think it very likely is. perhaps higher than expectations, then like it's going to get revenue on that continuously. Like in the many, you know, many, many years into the 2030s, even if this is a giant bubble that, you know, that, that does burst, you know, in this year, this year or next. That being said, they are, they do trade at by far the highest valuations in the semi. So how do you, how do you think of these, the semi cap names? They're like literally five. I mean, the, the two cannot be the same. I don't think Nvidia can be like such a low multiple and then semi cap, which has always been the low multiple guy.

be at such a high multiple, so there must be some sort of mean reversion at some point. Things have obviously run up a lot, like I said, because there's been maybe two decades of a lack of investment in any of these companies. No one cared about any of these companies for the last 20 years. It was all about Salesforce and Adobe and all this Twilio or whatever. That was sexy. And like, uh, Splunk, right? Yeah. The companies that their names were in some instances were just ridiculous. Like you should not be naming company that didn't matter because they were capital efficient, right? I was printing money. Yeah. Like now, like applied materials or ASML, like, like they need to spend millions of dollars building out more fab, uh, space and construction. Yeah. Applied materials is like super busy, but obviously, uh, I feel like there is some sort of. quasi cyclicity when it comes to these types of names and people need to be cognizant of that. There will be, at some point, some sort of stasis and overbuild. Trees don't grow to the sky. And I think maybe this generation of younger investors have forgotten that and remember what you're investing in and why and what your long-term thesis is. I think that's important. Yes. I think we can safely say that the 30 to

100% growth or over 100% growth in some like micron like that literally cannot continue for the next 50 years like I'd say the chance of that happening is zero um of course people have predicted that the Semi cycle would turn for the past two years and it hasn't happened yet What are you going to be looking for that when you see a you see b you see c you say hmm Ben this is kind of looking like the cycle is turning and Maybe I am either going to sell my semi stocks or definitely not buy more and just just hold out and go into protection mode like a little porcupine because this is the cycle of strength. What are you going to be? What is ABCD? Yeah, BAP research for my institutional subscribers. I'm tracking GPU capacity, GPU rental rates, gigawatts of construction planned. Obviously looking at Taiwan. What are the contract manufacturers building? What is their growth? Also looking at obviously the bellwether in video, what is your roadmap? How is that tracking? And then looking at the frontier labs, Anthropic, OpenAI, what are they building? How are they growing? How much more computer they taking? Are they growing headcount? And you kind of look at all those different things and you try your best to triangulate where you are in this cycle. I don't know, but.

you know, based on my research and industry conversations and speaking of customers, it feels like the AI hasn't penetrated as much as everyone thinks it has. And I just saw the Anthropic just leased 160,000 square feet on Hudson Street in New York for their New York headquarters. You know, they're going to put a thousand employees there. Like, so San Francisco, New York, where you are, are going to be the too big AI hubs will be there. So you get to say hi to them. Yes, I do. And the AI has not penetrated as much as people think. Is that bearish or bullish? Just be clear. I think it's bullish. Like I said, chat GPT was November 2022. We're hopefully going four years in this November. I mean, and this isn't like the iPhone where You can scale production and just chip a bunch of stuff to Apple stores and people just pick them up. This is more intricate. There's more parties involved. There's fewer bigger pocket buyers. And there's planning and roadmaps involved. So I think the race for computers is still there. And like I said, classical computing died. And what Jensen is saying, it's like the era of HPC. And there's still time. to really grow and go in that direction until God knows when you can do data centers in space, which is something for another time. Yes, that is a really key point that the semiconductor supply

chain is so complicated. And I'm not even saying that this is the hardest thing, but just to give people a sense, there's a company like Atomic Layer Deposition, ASM, they used to own ASML, I think, and they literally are putting layers onto the chip that are one atom thick or less than one atom thick. So like I think that it's really, really, really hard to do. And there are certain false bear signals that people could see of like, oh, shipments have flattened that actually are a sign of a constraint or a shortage rather than a lack of demand. Yeah. I mean, there's a lot of that going on. I tried not to. There's a lot of noise. Just look for signal. Look for the productivity. Look for the demand. Look how you're using these AI models, how you're getting, or it's not allowing you to use it. I think those are better signals where we are until more capacity comes online. And you see more of this AI actually diffuse into your everyday life. I think that's where a lot of people are missing. A lot of X is concentrated in tech and obviously in the Bay Area. So over there.

everybody is doing AI in Los Angeles, nothing, right? New York, maybe a little bit, right? Is the bodega down the street using AI? Probably not. So there's a way to really, there's a way, there's a ways to go to really see it permeate in the rest of society. And I think it will be a net benefit if you, if used in the right way for the right purposes. What are you following in terms of the frontier labs and their monetization? in terms of anthropic, in terms of open AI, what are you seeing in terms of their revenue? And just how fast is that growth relative to that revenue? And how does that compare to how their costs scale as well? The beauty of this business, obviously it's very capital intensive. Almost think about it as it's almost like a bakery type operation. You build this huge bakery plant, which is like the data center, and your inputs are... flour and water, or in this case, GPUs and electricity. And then your output, the tokens is the bread, right? But the flour and water cost a lot of money. And the biggest expense for building a data center is GPUs. The biggest expense for running a data center from a gap perspective is GPU depreciation. I think these GPUs are actually lasting longer than people expected, I think. You have H100s and A100s that are like over three years old still running and they're still useful. Not so much for training, but you can use them for inference. So useful life is still very, very there. But the thing is, because there's so much money going into these training runs, you make your money off the inference and the gross margin on inference

with the API pricing seems like it's trending towards, you know, the high 70, maybe even 80% gross margin. So this is like the new basically software paradigm. The new SaaS is inference, right? The question is, is if you're a software company, are you buying Tolkien's wholesale and adding your intelligence and selling retail? Do you have enough margin to survive against these frontier model companies that are coming after your business, who own the whole stack, who control their whole margin profile? I mean, that's why you've seen this huge derating in software like software was unassailable. Yes. Until now. I wonder why. But then the core point that software had very high gross margins that you would spend a lot of capital, hire a lot of expensive software engineers to make, pay software sellers a lot of money to make, to sell the software, distribute the software. But that once it was running, it was very, very high gross profit margins. Delivering AI does have lower margins. Delivering AI does have lower margins. I mean, not from what I've been reading.

The profitability on inference is there. I think, again, going back to the bread or oil analogy, I think if a company like Anthropic is made truffle-infused gold-flaked bread at a low cost and is selling pieces, thousands of pieces at a high price, then they're really making good money and they're probably heading towards cash flow positive at that. I mean, the guys, you said in February, they, they ate extra revenue plan. I mean, that's crazy. It's like the fastest growing company in the world at this point. So yeah, I just, I just pulled up sales force, a classic SaaS business. Yeah. Gross margins of like 72 to 77%. You're, so you're saying that, and you've read or heard that Anthropic's gross margins for inference are, are what you just want to be clear. I think in the seventies or higher. Yeah. That, that, um, that is pretty, pretty high. What about open AI? I haven't seen their numbers. I mean, from a user perspective, I do use some chat GPT and things like that, but for deep commercial work, it's mostly Claude and Fable and all this other stuff. I think they've really won on the enterprise, but I don't know what open AI is trying to do. Maybe 5.6. Is it? I know they have way more compute than an Anthropic, that's for sure. Do you think that... companies are happy with the money that they're spending. I know you watched Alex Karp on CNBC just absolutely trashing. He was very stern that the executives in corporate America are displeased with the cost of AI, the value that it provides in terms of outputs, and also the data lake that like Anthropic could be like

taking the data or the companies aren't allowed to own their own data setting that aside and then you know the CNBC people were saying Well, what are you saying? And Alex Karp said I am a vessel. I am speaking for corporate America It was also pretty pretty entertaining TV But that yeah, what do you what do you make of that argument that the companies are like whoa? I'm spending a billion dollars. This is ridiculous. Yeah, Alex is uh, I love him. He's like my spirit animals I think he just says whatever's on his mind and He doesn't care. And I think that's the way people should be. And that's really how things are being done. Even at NVIDIA, I posted a few times, when I was there in March in GTC, a VP of AI, Abdel Halak, was like, we use Opus to basically orchestrate. And we use Nemotron, which is our own open source model. And we can achieve frontier level results by combining these two. And we own our own compute.

So all those, the grunt work of tokens is owned by us and we don't pay anybody for it. So you, so the idea of using frontier models for everything is that's like when you blow your token allowance and you spend too much money on wasted projects and or like with Amazon, someone on Amazon spent like a million dollars or something on tokens. It just, it doesn't work. And I think that's where. Enterprises are heading with this like router approach where if it's really hard, use a frontier. If it's not, use a local model and have the frontier orchestrated. Does that make sense? That does make sense. And so short term that may lead to less demand for frontier tokens, but longer term, you think it actually is, is bullish because at the end of the day, the frontier is always going to be like pushing the limits and it's sort of. maybe like the luxury car, maybe it's like the Ferrari of the market. There's always going to be buyers for that who want the speed and they want to be the best. But maybe some people don't want that, or maybe some people want to mix. I mean, that's what I'm saying. Like the AI market is sort of maturing. Not one size fits all. There's different ways to do, like for, say you're like a CPG business and you want to have like a chatbot about like your your consumer products, just load all the data and then train it with like a open source model of the weights and everything. And that's it. Like talk about the product. It's not going to like solve math or find new chemicals or whatever for you. It's just going to be talking about, Hey, yeah, the frosted flakes has this much sugar in it. There you go. Whatever. Like, like the different parts of your stack are going to use different types of

models. It's overkill to use something like a chainsaw for everything when you just need like a butter knife. Right. Yeah. You don't need Albert Einstein for you doing data entry or honestly, even sending emails that aren't that important. Ben, what do you make of open source? This is the another bare thesis. Okay. Yes. AI is transformative, but open source models that charge so much less. So, so, so much less are just going to sap the pricing power. of the American frontier models. What's your reaction to that, number one? And where does NVIDIA's Nimatron come into this? Yeah, I think you need both. It's just not like I said, again, it's not one or the other. It's a combination of the two. And what I've told people is like, with NVIDIA, you get like a free open AR anthropic. Like, Nimatron is just as good. Like, they have like, you should look at the team of researchers, like the guys like, oh, I just got hired from

meta superintelligence because I'm working on nematron to solve science and hard math problems and the genetic workflow. So I was like, whoa, like, these are the people that Nvidia is hiring. Part of doing all these things is to keep your customers honest, right? Like, what if a company like Anthropic or so smart, they're like, oh, you know what, we'll just make our own Nvidia, we'll make our own chips, we'll make our own models. Like, we're just going to, we'll design you on everything. So like, if that happens, like, what, what leg does Nvidia have to stand on? They need to have their own model, right? So this is also part business, it's a business decision also to make sure your customers don't put you out of business, right? And I think a lot of the world wants open source. I think like the parallel again, going back to Apple frontier models are sort of like iOS, right? And it's sort of like closed and the open source is like Android. It can be anywhere and you can use it on anything. But the cool thing is that Nemotron in itself is optimized on CUDA and CUDA with the software layer is optimized for Nvidia hardware. So like you have basically developers all around the world on an open source contributing, figuring out ways for free to make this product better for you. You don't have to do anything. You can wake up like, oh, look, the community came up with this. And look, we, we made this a little bit better. Oh, that's awesome. Like I get free software updates. That's great. because I bought this and I have this. So that's like the benefit of open source and the silicon that it comes with. So that's good to know about NVIDIA has this ecosystem. I didn't know this Nemotron. That is

really good to know. People should look into that. And you know, if it's not obvious here, your giant NVIDIA bull, which is obviously worked out well. But Ben, just go back to open source. I'm let's say, there's a company's five people and they are going to do AI agentic workflows for coding. that really are going to transform their business and provide a million dollars worth of value. The problem is that they are that, you know, anthropic or open AI is going to charge them like four or 500 K to do that. So it's still a good deal, but it's really expensive. Then comes in Quinn, comes in deep seek, comes in GPU, all these Chinese models that instead of charging 500,000 or four or 500,000 are going to charge 6,000. How is that? Number one, are my priors wrong? Like, is that just not accurate? Or number two, how is that not wildly bearish for the Western AI things that are closed source? But where is your compute? Do you have your own compute and you're running your models locally or you're running these models hosted on something else and trying to get it to work? And the question is, will China block open source models out of their country? And then

If you're building on a model that's like blocked and you're halfway through building in your business, then what happens? What kind of business risk are you introducing to your company? So the question is, is like, how do you build your business in a way where you're the CIO or the CTO and you're not so model dependent on one company where tomorrow you're building on some sort of volcano and they erupt, pull your access or increase your pricing? What do you have to fall back on? by then they have you by the balls and then you're fucked. Okay. So, so people are going to pay a ton of money for Western models rather than Chinese models because they're worried about the Chinese government and China taking their data. That's what you're saying. That or at the same time, you just don't know where that's going to head. I think with, with anything with, obviously in China, I mean, like their playbook has always been, let's deflate the cost out of anything that we make and just try to export it around the world. And what I've coined in other segments, it's called the influence through influence. So the idea is like what China has been doing with companies in countries in Southeast Asia, like frontier Africa is like, like we don't have any running water, we don't have an airport, we have no infrastructure. Don't worry, China will come in, we'll build it for you and we'll lend you the money and all these natural resources that you get out of the ground and



these rare herbs, we'll take it. Thanks. So having the next thing China is going to do is like, you know what? Oh, you're in Zimbabwe. You want AI? No problem. We'll build you a Huawei data center and we'll give you the model surround on it. And it's the GLM or deep seek. And by the way, if you look up Tiananmen Square, it never happened. And they basically influence. This is like a global policy perspective from America and democracy, right? how the world views America through the lens of China and these models. So you're influencing through inference. And I think that's another thing that America can afford to lose on the race for AI of the world. So your argument that China is not going to displace the pricing power of these Western models is based on geopolitics rather than tech. I am not saying that you're wrong and that totally like the reason that people are all around the world and Europe use Microsoft as opposed to like. you know, some technology developed in Singapore is precisely for that reason or just that we have kind of network effects in the American tech stack. But don't you think with the vast sums that are being spent and will be spent by companies on AI that, you know, if it really is so transformative, wouldn't you want to save 2 billion a year? Like if you are, I don't know, Coca-Cola or something. And you know, in many years and you're spending, let's say 50 million a year as Coca-Cola, wouldn't you want to

Cut that to 15 million. Are you heavily, heavily incentivized to do that? Sometimes there's career risk with putting all your eggs in one basket. And I do see somewhat of a contingent hybrid approach when it comes to model selection by CIOs and CEOs of companies. You can't go all in on one. You have to have like a few that you're basically routing around and then you'll have like a blended lower cost. Hopefully. Ben, what do you think is the most overrated stock or segment of the supply chain? Overrated. I suspect that you're not short any semiconductor stocks, but if you were running a long short hedge fund, and you had to be short, which ones would you be and why? I think there's a lot of these companies that are doing optical materials, like AXTI or indium phosphate. Again. You're just going after a raw material that's supposedly very high in demand, that will have boom and busts and things like that. To me, it's like, I like to be in places that have some sort of defensible mode as good management that will last over time. And those types of names to me are just very esoteric. Obviously, there are some cartoon characters on X who shall not be named. that have

large followings who tell people to go into these names, which to their own risk, they can do all that stuff. Yes. You know, I understand that there are some commodities that actually are very rare, like the photo resists and they're very hard to make and like two to four Japanese companies make them. But yeah, I mean, like that makes sense to me that why if the commodity is literally in the periodic table, that sounds like something that should not have. Pricing power in the long term. Correct. I mean the whole idea is like the idea of modes and how hard is it? I think that's where You know, I'd live through it with LED lighting, right? It was like a commodity. It's like a diode and we had competitors from China and Korea and like everyone was doing it in Amazon So it's like if you're getting squeezed from all different directions like that's not a good feeling to be in because everyone's chasing the same market you want to be in something that's very hard and has definitive layers against it. And I think when you look at, from a macro perspective, like I said, all hardware eventually commoditizes, right? The last scalable consumer electronic device that hasn't commoditized is what you're talking to right now, your MacBook, your iPhone, right? How come it's just a computer, right? It's just like a phone. Like why hasn't it, why is it a thousand dollars? How are they getting these like, because the software layer matters?

The co-design matters, right? And I would say brand, which I think in the semiconductor world, brand kind of, from a consumer's perspective, doesn't really matter. Like no one is like, oh, I'm going to buy this phone because it has this particular type of NPU. It's not brand, but the thing is, you know an iPhone has an operation of four nines. It's going to work 99.99% of the time, uptime, versus like these like weird phones that were coming out. You had to reset it a bunch of times and it was like an iPhone wannabe. You know this phone works, dude. So you're paying for reliability, the same thing with Nvidia. You're going to get four nines. And if something goes wrong, someone's there to fix it. You're paying for that. You're paying for that certainty. But if you want to take the risk and make your own chip or buy from AMD or one of these startups, go for it. Be a guinea pig. See where that takes you. You'll save. It will save money on the front end where you're going to pay on the back end. I've lived through this many times, you know, buying the cheapest thing. And on the back end, I pay for it. So I like, I'm done. I rather just, hey, you want to make this

margin? Okay, I'll pay for it. And I just move on. Ben, obviously I'm not going to ask you to give, you know, all of your alpha away for free, but could you give us one name or sector that you think is underappreciated in the semiconductor supply chain right now? as was perhaps like, you know, you wrote about blue energy in the winter of last year. I wrote about this and people are going to give me a lot of crap. This company called super micro. I think it was like a Soutrini favorite. I think it was like 600 bucks. And then like they managed to burn through all their cash and then smuggle the GPUs to China. I don't know why the guy was doing that. But like It's a real company in San Jose. They have people working, building these server racks, and their NVIDIA needs them to survive. AMD needs them to survive because they make a good, liquid-cooled product. They can make AI racks and hardware, right? Dell, obviously, I love Michael Dell. He's like the American dream guy out of college, out of his door room. I love him, right? And you have companies like HPE doing the same thing. I think Supermicro just had maybe some bad family members that were doing wrong things, but I think from a valuation standpoint and the IP that they have, and obviously they just raised about 7 billion of dilutive funding recently. I think JP Morgan led it. If people are putting 7 billion into this company and it's probably we'll figure out what institutions did, then there's an order book behind it. And if SpaceX.

just raised 85 billion. They're going to be adding more capacity to Colossus in Memphis. And Supermicro is going to get that call. I don't know what their gross margin is going to be. It's going to be low, but hopefully they can squeeze a few points out of it and really rebuild this business of what was and get back to their glory days. And probably not 600 a share, but something reasonable. And I think that from a fallen angel governance perspective. Uh, needs to be, you know, looked at again. That is interesting. I remember when, when, uh, she talks about it on my show and it had an accounting scandal then, and then it, you had huge demand from AI. It's went up a tremendous amount and it's not as capital expensive. There was some bad thing that people should look into. Um, but yeah, okay. That's interesting. Ben, one question I wanted to ask you, what do you make of the bear case for. The software in the semiconductor supply chain and not talking about NVIDIA, I guess, although I kind of am maybe, but I'm in particular to the two companies are cadence and that they to explain for our

audience, they design the software that the engineers are going to use when they make the design the chips. And this has been the ultimately like high power software and like a lot of software, super high retention. And then they, they get, you know, net dollar expansion upsells. Yada, yada, yada, so great business. But I mean, if AI really is so transformational, then maybe the AI companies are going to design their own software and save money on the, you know, hundreds of millions of dollars that it probably costs to, to license this software. So, you know, this is probably the stocks that are up at the least this year, cadence and synopsis. What's your outlook on this on these on these two names and the overall the bear case on semi software that like AI can make software. Calls to software is zero. Yeah. I mean, that's, I think when open AI released the jalapeno chip, they said they did it themselves. I mean, I don't know the details of what they use, but EDA, Electronic Design Automation Synopsis and Cadence have been a duopoly for a long time and they were sort of unassailable. And obviously now with software multiple headwinds, people are wondering if AI companies can just do it themselves. I think the jury's still out. You still get a lot of libraries and fies and things like that that these companies offer. And if Jensen put five, was it 5 billion or 2 billion? How much do you put in synopsis? Two billion. Two billion. Okay. That's two billion more dollars than I ever put into the synopsis. So, I mean, I have a lot of respect for him. NVIDIA is not a company that

like to spend money on nothing. I mean, I was there. I remember people wrote articles about it. I'm like, oh, there's no free food, but the, the burrito bowl was like four bucks and I had to pay 5 for a cold brew and there was only free black coffee. Like it's not like meta. It's not like Google with like fancy. You're not getting your massages. Yeah. None of that. Like these people are like grinding. No one was like, remember I was there? I was out there at like 1250 and I was like, I was checking the stock price. No one was on their phone looking at the stock price. Everyone's just happy to be there. Kid brought his daughter to work. There's no parking in the parking lot. They're expanding another facility. Mission is the boss. They're focused on the mission. And when I speak to some people that even worked there for 20 years, I'm like, dude, you're worth probably way more than me. Why are you still here? He's like, this is my moment. I'm going to retire and stay home. I'm getting fights with my wife. Like you want me to

like, we designed this company for accelerated computing and we're doing it now. You want me to like just stop? I'm like, I guess not. So coming back to EDA. Yeah. I mean, the jury's still out. We don't know. I don't want to bet against Jensen, but at the same time, I think where the investment and synopsis and I think the salt systems also, they have a partnership. The idea is how do you use the libraries and the models to model the physical world? into the digital world, omniverse. And if you can model the real world into digital world at scale with compute, then you can solve a lot of problems and physical things digitally before you have to actually do it physically. And that in itself is a big money saver. That's how they're designing all these data centers and products and things like that. Everything is virtual rather than physically trying to do trial and error. So the idea of doing things physically trial and error and wasting money and waste and time and virtually doing it with EDA tools. I think there's a net benefits of society and people are so early and they haven't seen it. There's also medical devices. It's not just semiconductors, right? They do a lot of things on the design front. And what about the IP licensing business that Synopsys and to a lesser degree, Cadence have? The big leader in this, of course, is ARM that IPO'd like two years majority owned by SoftBank. And then the other players, I guess, are Rambus. But but I've heard this bear argument on IP licensing like Nvidia, they have all these geniuses working in Nvidia. They're not going to be paying synopsis or cadence or Rambus or ARM to do all this stuff. I mean, maybe they will be paying ARM, but that's a different story. But what do you and at the same time, if that argument is wrong, the royalty type business has a gross profit margin.

quite close to a hundred percent. That's an extremely good business. From a business perspective, it would be weird if you invested in the company and then all of a sudden you try not being their customer anymore. It kind of hurting your own investment. Yeah. So NVIDIA invested in Synopsys and, um, former CEO of Intel now Lipitan, former CEO of Cadence. So just saying is, you know, is he going to cancel the contracts? I don't know. Ben, final question. Tell me about capacitors, specifically MLCC's multi-layer ceramic chip capacitors. Investors, whether they're, you know, individual investors, retail investors, specialists or institutional investors, the big banks are now writing reports about this are saying the new memory cycle. It was out of the golden or JP Morgan that said the new memory, the new

bottleneck is in MLCCs, not memory. So these would be stocks like Vichay comes to mind, which I actually think Satrini said on the show like three, three years ago, but these stocks are certainly catching a bid to put it mildly. Is this hype or is this, there's something there, there. Yeah. Again, chasing capacitors and resistors, which are like on the low end of semiconductors, which like a business that I never wanted to be in, that tomorrow some company out of China will just pop up and be funded by the government and flood the market with a resistor capacitor. Like that's not a business you want to be in. But again, people love this notion of chasing bottlenecks. I think that will probably end in tears at some point. Not for me. Yeah. Well, actually Vichay actually is down 35% from the peak earlier in late, late June. Okay. But Ben, that whole argument, I'm not saying you're wrong, but people said that. But we went through it. If you go back in time in history, back to the, the iPhone super cycle from 2008 or 2007 to 2010, like when I was building, uh, I think LED products and like we couldn't get power supplies because all the capacitors were in shortage because the. Apple was eating up the world's capacitors and resistors for their iPhones, right? And the same thing is sort of happening now. You have maybe it's bifurcated. You have a new iPhone coming out and you have this existing iPhone cycle of people of like the Stasis of capacitor demand. And then if you look at the the board with the GPUs on it, there's like a million of these like barrel looking things which are the capacitors on it and Obviously they're using a lot and Nvidia selling more racks. So then you see this spike up in demand that they haven't been able to catch up, but eventually will catch up. So it's like, it's a moment in time before they just catch up. You don't think they're going to, they want to sell more capacitors. They will, but there's just a lag to catch up to the demand. Everything you said is true, but it also is a true argument about memory and yet you're bullish on memory. What's the difference? Think of the memory, obviously there's.

three players and then the HBM, which I've written about it. I've been with memory, which requires a lot more wafers. Yeah, it's bad. There's like wafers away. So then like it's it's almost like making a lasagna that you have to like stack multiple layers exactly on top of each other. And then the wires that go through that connect all these stacks of memory. have to be like nanometers precise to make sure they basically all connect and then land on the board, okay? So it's not like the memory that goes like you

would go to Fry's electronics or computer world where you like stick it in like I got some crucial memory. It's all good Like this is this is a little different like when you're stacking layers like that and Not all of them go perfect every time and you throw away a little bit and you have your yield isn't there, that's where it gets a little special and challenging. Maybe at some point it'll make like a robot that can fully do it and it's like 99% accurate, but until then it's a little harder and then I think they wanted to do, they wanted to go up, I think the highest is like 14, they wanted to go to 16 and they weren't able to do it. I didn't know that extra layer was just too hard, aligning everything. So there you go. It's like, that's the difficult part at this point. When do you think memory is going to come online into efficient scale to cause prices to go down? Prior early 2028. So you'll probably see something mid 2027 where people like freak out. Freak out because memory prices are going to go down. Yeah. I see. I see. Everyone's basically like memory is like this. There's like three doors and then everyone's just going to try to like rush out of these three doors and you'll see like this huge. like stampede I feel like because by then it's like okay like we we've done all the catbacks we built the fabs they're online we're running 24 seven you know three shifts like we're doing it we caught up like at that point like what else what's holding it back we've done everything the market is flooded okay so then than what? Yes. I mean, Micron and SK Hynix Samsung almost guaranteed to just be printing absolute hundreds of billions of dollars in profit, which is a ridiculous thing to say over the next year. But it is it is true that generally the most money is made in investing buying before that actually happens and that if you buy it like a company that they are making tons of money, but their pricing parties gradually going down. And so

you know, they're like, they're still very profitable, but it's just the rate of change isn't very good. That's a far less exciting opportunity. Yeah, you're a good study of history. That makes sense. Ben, I love your work. Tell us about where, what kind of research do you do on Semi's AI? Who is it for? And what should people expect when they check it out? Yeah, if you can find me at BEP Research or Benitos on X and I really try to be a big system lover thinker. So I connect the silicon to the software to the models and it's been like a really, it's been like a closed loop how the models and the software are driving the hardware and it's sort of like recursive and I'm

looking at ways how AI is being applied. I think the next inflection is obviously the sciences and what David Friedberg was saying on All In and how Anthropic is trying to use your data or what Alex Karp was saying. I think that will probably be the next inflection for applied AI and obviously robotics. And my audience is mostly high-network family offices investors and some institutional funds which I consult with on deeper thought pieces and research work for them. on a project basis. So overall, it's been pretty exciting. I like meeting people in industry, going to conferences, learning about new technologies and new things. I generally enjoy this stuff. It's exciting to me. I get to use my degree. And at the same time, it's something I'm comfortable talking about. I don't get into the weeds about how many dies some NVIDIA chip has or like this glass substrate is missing or whatever it is, I think. you need to see the forest through the trees and see the bigger picture and connect all the dots to make a thesis. And as an investor, your job is to make sure your thesis is correct because when you hold a position, you're choosing to buy that position every day, right? So I think having that framework and that mindset helps you figure out what your directories or things you're headed in with those investments. Thank you, Ben. You're welcome.